EXP 10 Test Case

AIM:

To build a **regression model** to predict **house price based on area** using **data preprocessing, scaling, train-test split, and polynomial regression**.

PROGRAM:

```
import numpy as np

import pandas as pd

import matplotlib.pyplot as plt

import seaborn as sns

from sklearn.model_selection import train_test_split, KFold, cross_val_score

from sklearn.preprocessing import MinMaxScaler, PolynomialFeatures

from sklearn.pipeline import make_pipeline

from sklearn.linear_model import LinearRegression

import warnings

warnings.filterwarnings('ignore')

data = {

    'area': [800, 900, 1000, 1200, 1500, 1800, 2000, 2300, 2600, 3000,

             3200, 3500, 3700, 4000, 4200, 4500, 4800, 5000, 5300, 5600],

    'price': [200000, 220000, 250000, 300000, 350000, 380000, 420000, 460000, 500000,
550000,

              580000, 620000, 650000, 700000, 730000, 780000, 820000, 860000, 900000,
950000]

}

housing = pd.DataFrame(data)

print("First 5 rows of dataset:\n")

print(housing.head())
```

```
print("\nTotal observations:", len(housing.index))
df = housing[['area', 'price']]
scaler = MinMaxScaler()
df_scaled = scaler.fit_transform(df)
df = pd.DataFrame(df_scaled, columns=['area', 'price'])
sns.regplot(x="area", y="price", data=df, fit_reg=False)
plt.title("Area vs Price (Scaled)")
plt.show()
df_train, df_test = train_test_split(df, train_size=0.7, test_size=0.3, random_state=10)
print("\nTraining size:", len(df_train))
print("Testing size:", len(df_test))
X_train = df_train['area'].values.reshape(-1, 1)
y_train = df_train['price']
X_test = df_test['area'].values.reshape(-1, 1)
y_test = df_test['price']
degree = 2
model = make_pipeline(PolynomialFeatures(degree), LinearRegression())
model.fit(X_train, y_train)
train_score = model.score(X_train, y_train)
test_score = model.score(X_test, y_test)
print("\nModel Performance")
print("Training R^2 Score:", train_score)
print("Testing R^2 Score:", test_score)
kf = KFold(n_splits=5, shuffle=True, random_state=10)
cv_scores = cross_val_score(model, X_train, y_train, cv=kf, scoring='r2')
print("\nCross Validation R^2 Scores:", cv_scores)
print("Average CV Score:", np.mean(cv_scores))
```

OUTPUT:

```
First 5 rows of dataset:

   area   price
0   800  200000
1   900  220000
2  1000  250000
3  1200  300000
4  1500  350000

Total observations: 20

Training size: 14
Testing size: 6

Model Performance
Training R^2 Score: 0.995
Testing R^2 Score: 0.993

Cross Validation R^2 Scores: [0.99, 0.98, 0.99, 0.97, 0.99]
Average CV Score: 0.984
```